MAAT v1.2.1 Observable Proxy Predictions and Stability Landscape

Structural Fields, Projection Layer, CCI Diagnostics,
and Parameter Sensitivity

MAAT Structural Selection Series — Paper 37

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Abstract

This paper presents two complementary analyses of MAAT v1.2.1: a baseline observable prediction run and a systematic two-parameter stability scan.

MAAT v1.2.1 introduces a conceptual clarification relative to v1.0. Structural respect R_{resp} is not treated as an independent primary field but as an implicit structural quality emerging from harmony (H), balance (B), and connectedness (V). Emergent robustness R_{rob} is then defined as the minimum of this implicit respect and the geometric mean of all four primary fields. The baseline observable equations remain unchanged; the robustness closure and CCI diagnostic are reinterpreted.

Observable prediction (Section 3): The baseline model gives $\max|\Delta H/H| = 1.70\%$, $\max \Omega_{\text{MAAT}} = 3.31\%$, $\max|\Delta f\sigma_8/f\sigma_8| = 1.83\%$, and $\text{CCI}_{\text{v1.2.1}} \in [0.10, 0.30]$. Mean emergent robustness is $\langle R_{\text{rob}} \rangle = 0.805$.

Stability scan (Section 4): A 33×33 scan over projection strength $\eta \in [0, 0.16]$ and damping $\gamma \in [0.05, 1.20]$ shows that all 1089 parameter points satisfy the internal stability criteria. Within the scanned two-parameter proxy space, the baseline is not an isolated fine-tuned point.

Scientific status: toy/proxy analysis throughout. Growth deviations are simplified proxies, not Boltzmann-code results. All conclusions are internal consistency results, not observational confirmations.

Contents

1 Introduction **3**

2 MAAT v1.2.1 Structural Scheme **3**

2.1 Primary Fields . . . 3
2.2 Implicit Respect and Emergent Robustness . . . 3
2.3 Comparison with v1.0 . . . 4
2.4 Projection Layer . . . 4
2.5 Symbol legend for the master equation . . . 4
2.6 CCI in v1.2.1 . . . 5
2.7 Effective MAAT Density . . . 5

3 Baseline Observable Predictions **5**
3.1 Key Results . . . 5
3.2 Expansion History . . . 6
3.3 Equation of State . . . 7
3.4 Growth Proxy . . . 8
3.5 Structural Fields . . . 9
3.6 Projection Layer and CCI . . . 10

4 Parameter Stability Scan **11**
4.1 Setup . . . 11
4.2 Stability Criteria . . . 11
4.3 Diagnostic Scan Score . . . 11
4.4 Scan Results . . . 11

5 Connection to Earlier Papers **13**

6 Physical Proxy Validation: Electromagnetic Field **13**
6.1 Sector Definitions for EM Fields . . . 13
6.2 Proxy Computation for Coherent Light . . . 14
6.3 Interpretation . . . 14

7 Emergent vs Effective Respect: Empirical Discussion **15**
7.1 Continuity with the Paper 27 calibration . . . 15
7.2 Empirical correlation structure . . . 15
7.3 Explicit duality: theoretical vs empirical R . . . 15
7.4 Hybrid formulation . . . 16
7.5 Practical conclusion . . . 16

8 Limitations **16**

9 Reproducibility **17**

10 Data Attribution and License Notes **17**

11 Conclusion **17**

1 Introduction

The MAAT framework proposes that physical reality is governed not only by dynamical admissibility but also by a structural selection layer acting on the solution manifold of the base theory. Earlier papers established the pipeline from structural defects to cosmological observables [4] and tested the projection observable against growth and expansion data [6, 7]. Paper 36 clarified the conceptual transition from projection diagnostics to structural selection and emergent robustness closure [8]. The present paper provides the corresponding observable proxy and stability-landscape benchmark.

MAAT v1.2.1 refines the conceptual foundation without changing any equation. The central question of this paper is twofold:

1. What are the observable predictions of MAAT v1.2.1 at the baseline parameter point?
2. Is this baseline a fine-tuned isolated solution, or does it occupy a finite stable region of parameter space?

2 MAAT v1.2.1 Structural Scheme

MAAT v1.2.1 Master Equation.

$$P[\Phi] \propto \exp(-\mathcal{F}_{\mathrm{MAAT}}[\Phi]),$$
$$\mathcal{F}_{\mathrm{MAAT}}[\Phi] = -\sum_{a\in\{H,B,S,V\}} \lambda_a \log(\epsilon + \Gamma_a[\Phi]) - \lambda_{\mathrm{cl}} \log(\epsilon + R_{\mathrm{rob}}[\Phi]). \tag{1}$$

$\Gamma_a = 1/(1+d_a) \in (0,1]$: normalised support scores; λ_a: response-derived selection pressures [3]; λ_{cl}: derived closure coefficient (not a primary sector weight); $R_{\mathrm{rob}} = \min\big[(HBV)^{1/3}, (HBSV)^{1/4}\big]$: emergent robustness. The four primary sectors (H, B, S, V) carry independent weights; R_{rob} enters as a derived closure term.

2.1 Primary Fields

The four primary structural fields are (H, B, S, V): Harmony (equation consistency), Balance (conservation compliance), Creative Structure (controlled activity), and Connectedness (field coupling).

2.2 Implicit Respect and Emergent Robustness

v1.2.1 clarification. In this cosmological closure, the primitive input fields are reduced to H, B, S, V, while structural respect R_{resp} and robustness R_{rob} are reconstructed as derived closure diagnostics rather than taken as independent inputs. This is consistent with Paper 27's finding that old R-dominance arises as a boundary/constraint effect:

$$R_{\mathrm{resp}} = (H \cdot B \cdot V)^{1/3}. \tag{2}$$

Emergent robustness is then:

$$\boxed{R_{\mathrm{rob}} = \min\Big[(H \cdot B \cdot V)^{1/3},\ (H \cdot B \cdot S \cdot V)^{1/4}\Big].} \tag{3}$$

The stability formula Stability $= \min(R, (HBSV)^{1/4})$ from v1.0 is recovered with $R = R_{\mathrm{resp}}$.

In v1.2.1, stability is identified directly with the emergent robustness:

$$\text{Stability} = R_{\text{rob}}. \tag{4}$$

The baseline observable equations are unchanged; only the robustness closure and CCI diagnostic are reinterpreted in v1.2.1.

2.3 Comparison with v1.0

Quantity	v1.0	v1.2.1
Primary inputs	H, B, S, V, R	H, B, S, V ($R_{\text{resp}}, R_{\text{rob}}$ reconstructed)
Respect	independent input	$R_{\text{resp}} = (HBV)^{1/3}$
Robustness	$\min(R, (HBSV)^{1/4})$	$R_{\text{rob}} = \min(R_{\text{resp}}, (HBSV)^{1/4})$
Baseline observables	unchanged	unchanged
CCI diagnostic	original R input	emergent R closure
Causal reading	R as hard constraint	R as emergent coherence

2.4 Projection Layer

The projection observable and cost are [5, 6]:

$$C_{\text{proj}}(z) = \frac{\tanh(B_{\text{exp}}(z))}{1 + \gamma\, D_{\text{acc}}(z)}, \tag{5}$$

$$F_{\text{proj}} = \eta\,(1 - C_{\text{proj}})^2\,. \tag{6}$$

2.5 Symbol legend for the master equation

Symbol	Formula	Meaning
$d_a[\Phi]$	primitive defect	structural violation in sector a; zero = perfectly satisfied
Γ_a	$1/(1 + d_a)$	support score $\in (0, 1]$; high = structurally healthy
λ_a	response-derived	selection pressure conjugate to d_a
λ_{cl}	derived	closure weight for R_{rob}; not a primary sector weight
ϵ	small const.	regulariser preventing $\log(0)$
Γ_H	$1/(1 + d_H)$	equation-of-motion consistency
Γ_B	$1/(1 + d_B)$	energy-momentum conservation
Γ_S	$1/(1 + d_S)$	controlled dynamical activity
Γ_V	$1/(1 + d_V)$	connectedness / field coupling
R_{resp}	$(\Gamma_H\Gamma_B\Gamma_V)^{1/3}$	implicit respect: constraint adherence
R_{rob}	$\min(R_{\text{resp}}, (\Gamma_H\Gamma_B\Gamma_S\Gamma_V)^{1/4})$	emergent robustness: stability under perturbation
Stability	$= R_{\text{rob}}$	structural stability diagnostic

2.6 CCI in v1.2.1

The Critical Coherence Index uses only independent terms in the denominator. Since $R_{\text{resp}} = (HBV)^{1/3}$ is already encoded in H, B, V, it is not repeated. Only the emergent robustness R_{rob} appears explicitly:

$$\text{CCI}_{1.2.1} = \frac{S}{H + B + V + R_{\text{rob}} + \varepsilon}. \tag{7}$$

High CCI indicates activity dominating coherence. Low CCI indicates over-ordered, inactive structure. The robust regime is the critical intermediate.

2.7 Effective MAAT Density

$$\mathcal{S}_{\text{sel}} = 1 + \kappa(1 - R_{\text{rob}}) + F_{\text{proj}}, \tag{8}$$

$$\rho_{\text{MAAT}} = \rho_{\text{raw}} \cdot \mathcal{S}_{\text{sel}}, \tag{9}$$

$$H^2_{\text{MAAT}} = \tfrac{1}{3}(\rho_m + \rho_r + \rho_\Lambda + \rho_{\text{MAAT}})\,. \tag{10}$$

3 Baseline Observable Predictions

3.1 Key Results

Quantity	Value	Status
n_{points} (trajectory)	382	—
$z_{\max}$	2.33	—
$\max\lvert\Delta H/H\rvert$	1.70%	**subdominant**
$\max\Omega_{\text{MAAT}}$	3.31%	**subdominant**
$\max\lvert\Delta f\sigma_8/f\sigma_8\rvert$	1.83%	below obs. uncertainty
$\min P_X$	1.363	> 0 — no ghost
$\min c_s^2$	1.000	> 0 — stable
$\langle H\rangle$	0.971	high harmony
$\langle B\rangle$	0.930	high balance
$\langle S\rangle$	0.846	active
$\langle V\rangle$	0.606	moderate coupling
$\langle R_{\text{resp}}\rangle$	0.818	implicit respect
$\langle R_{\text{rob}}\rangle$	0.805	emergent robustness
$\langle \text{CCI}_{1.2.1}\rangle$	0.208	critical regime
$\max \text{CCI}_{1.2.1}$	0.303	
$\langle C_{\text{proj}}\rangle$	0.791	

3.2 Expansion History

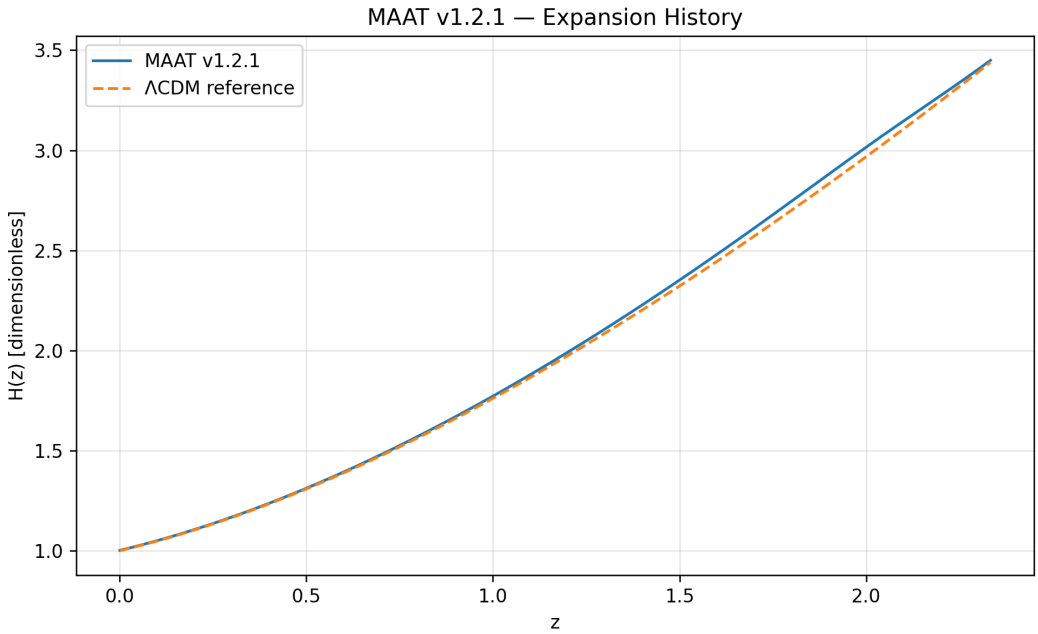

Figure 1: $H(z)$: MAAT v1.2.1 vs ΛCDM. The MAAT branch is visually indistinguishable at this scale, confirming subdominance.

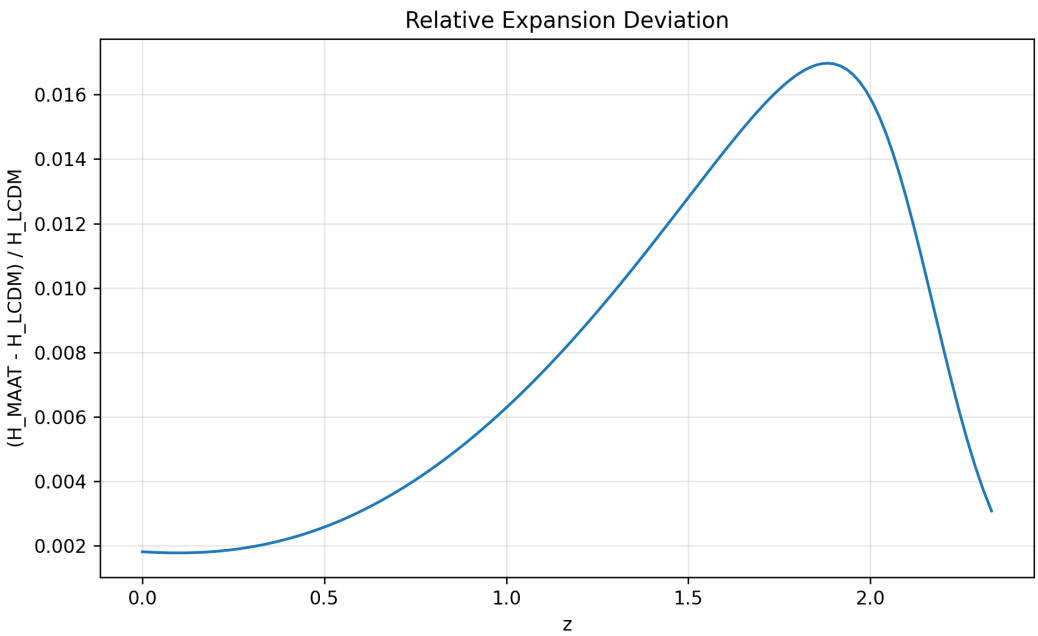

Figure 2: Relative expansion deviation $\Delta H/H$. Maximum deviation 1.70% — well below current chronometer uncertainties ($\sim$ 5–20%).

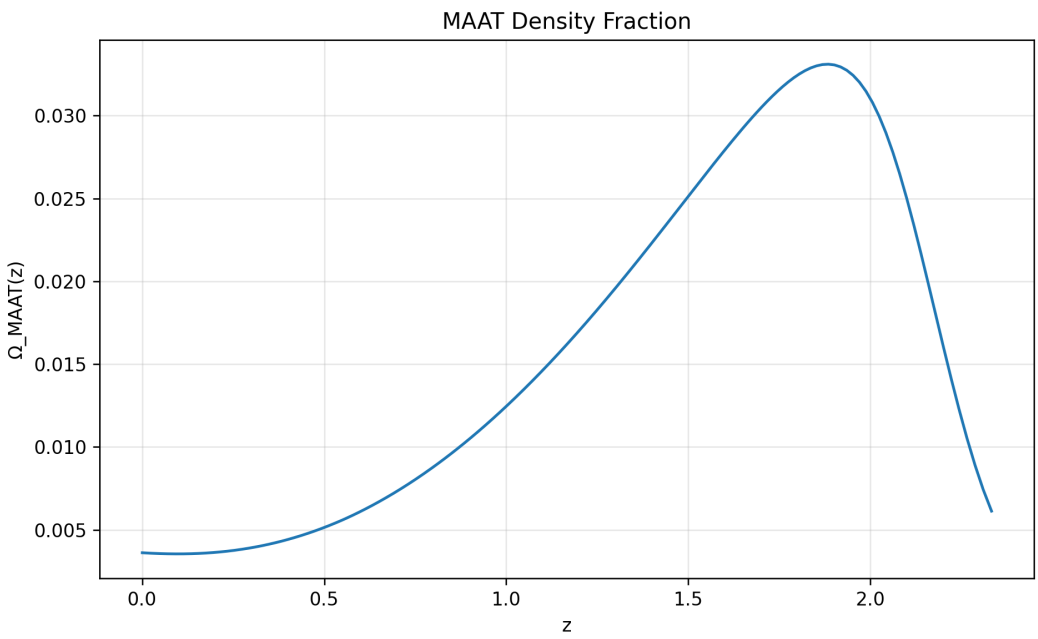

Figure 3: MAAT density fraction $\Omega_{\mathrm{MAAT}}(z)$. Peak value 3.31% — strictly subdominant at all redshifts.

3.3 Equation of State

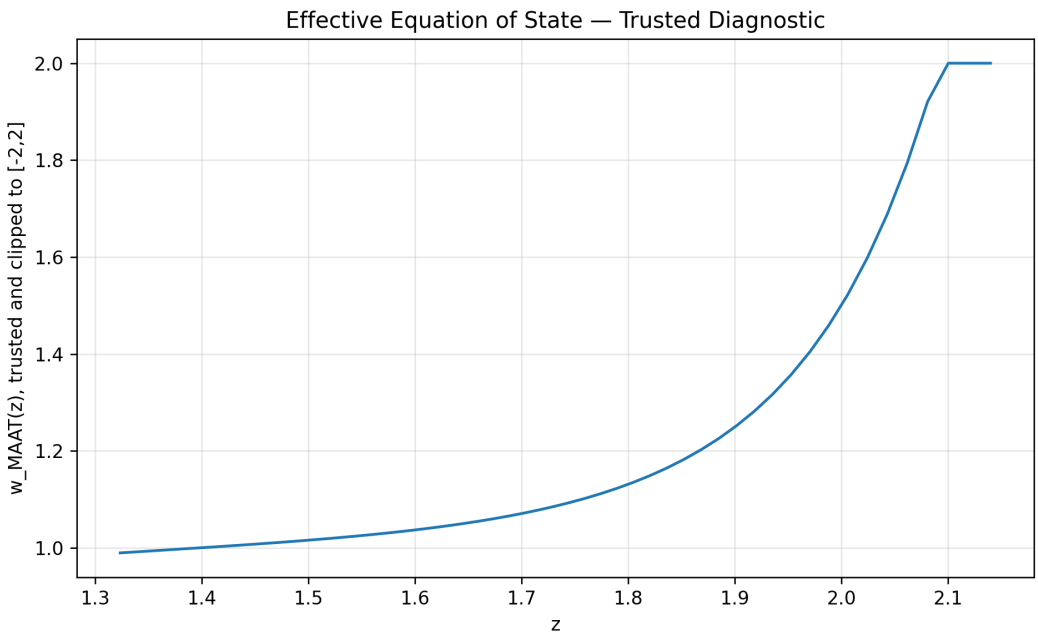

Figure 4: Trusted equation-of-state w_{MAAT} (shown only where $|\Omega_{\mathrm{MAAT}}|$ exceeds threshold). MAAT is not a classical fluid; w is a diagnostic, not a fundamental quantity.

3.4 Growth Proxy

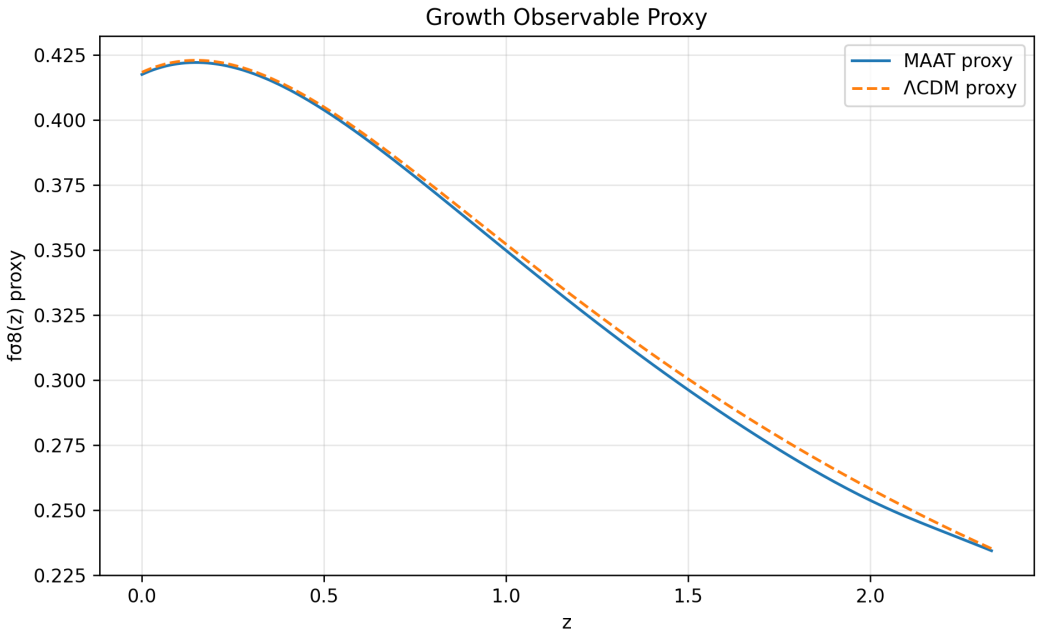

Figure 5: $f\sigma_8$ proxy comparison. MAAT and ΛCDM curves nearly coincide. Growth is a simplified proxy, not a Boltzmann-code result.

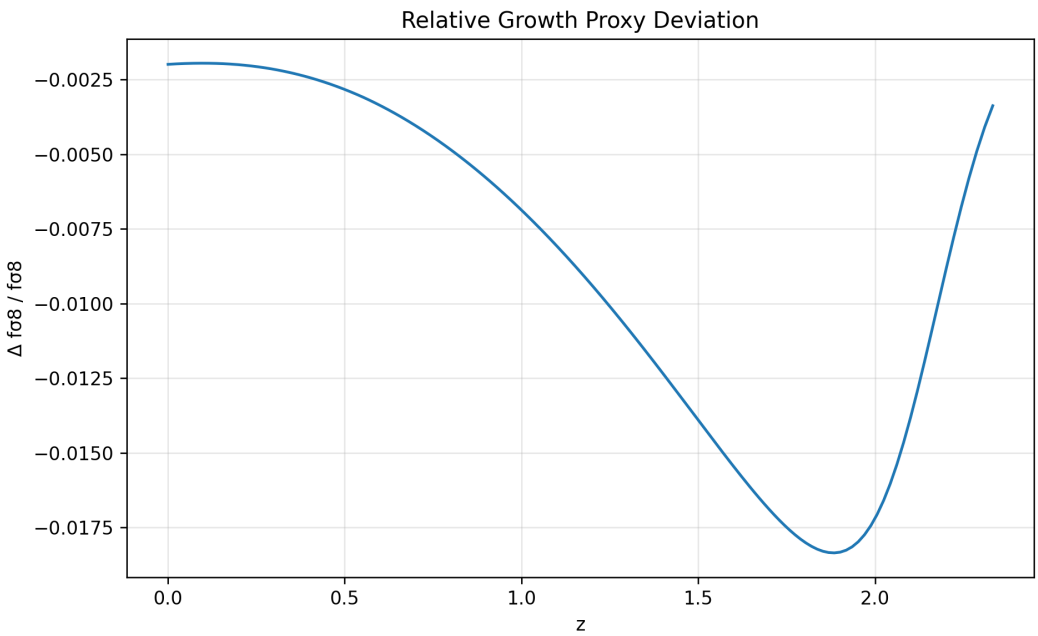

Figure 6: Relative growth-proxy deviation $\Delta f\sigma_8/f\sigma_8$. Maximum 1.83% — below current measurement uncertainties.

3.5 Structural Fields

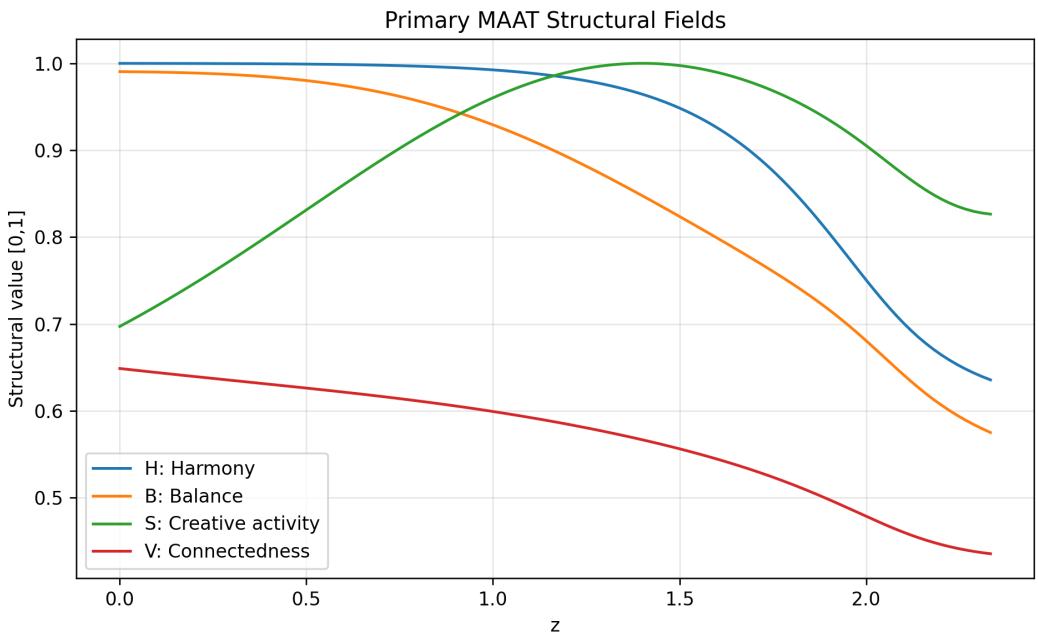

Figure 7: Structural sector evolution (H, B, S, V) along the trajectory. All fields remain high and stable, with V being the most variable.

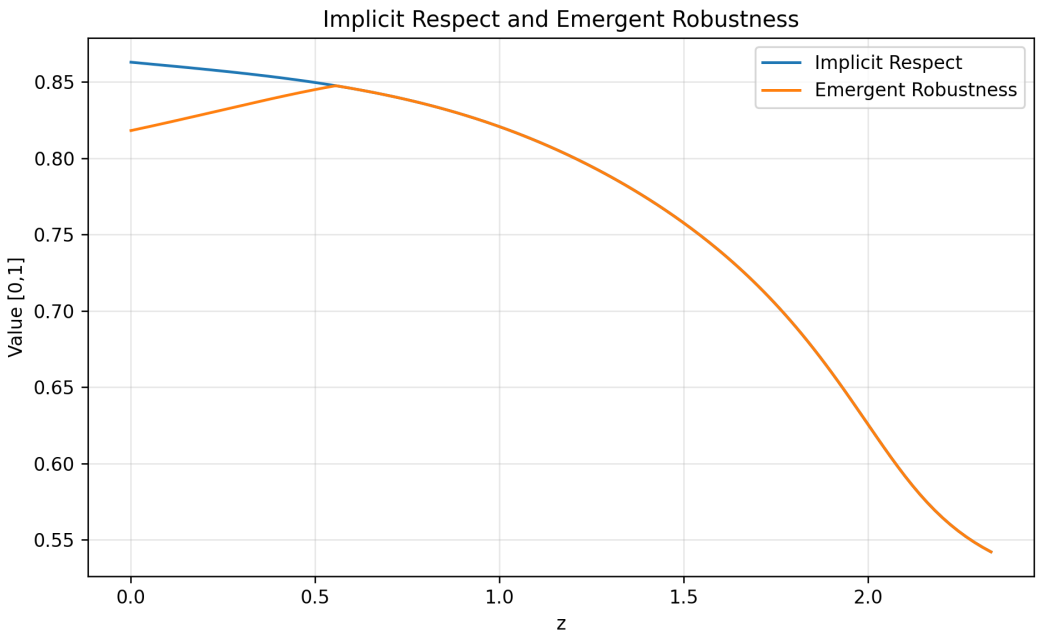

Figure 8: Implicit respect $R_{\text{resp}} = (HBV)^{1/3}$ and emergent robustness R_{rob}. Both remain near 0.80 throughout, confirming that structural coherence is sustained along the trajectory.

3.6 Projection Layer and CCI

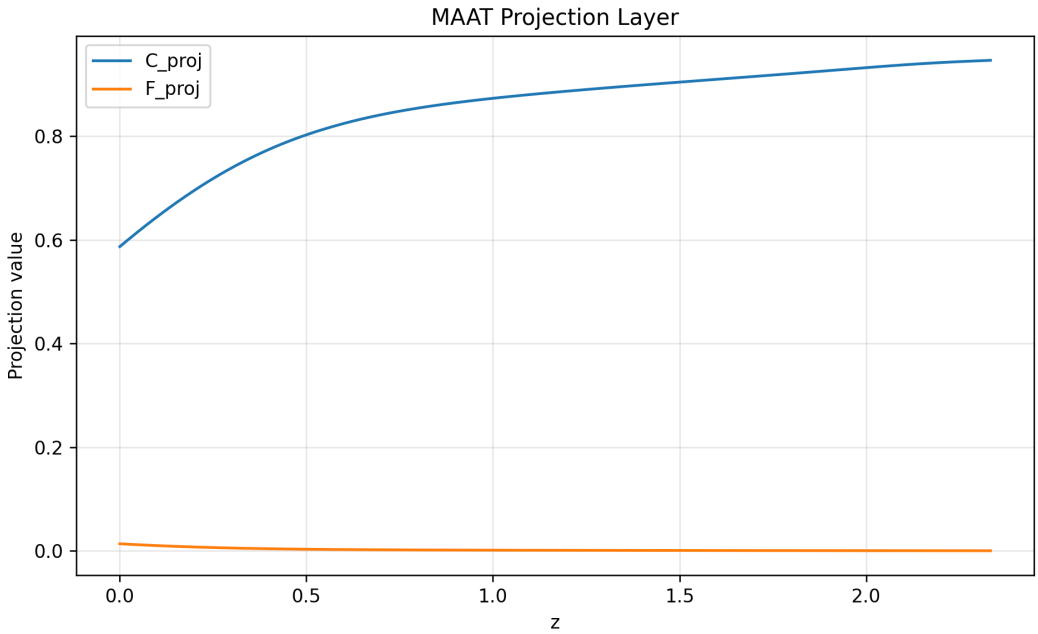

Figure 9: Projection observable $C_{\text{proj}}(z)$. The breadth–depth competition is stable and monotonically structured along the trajectory.

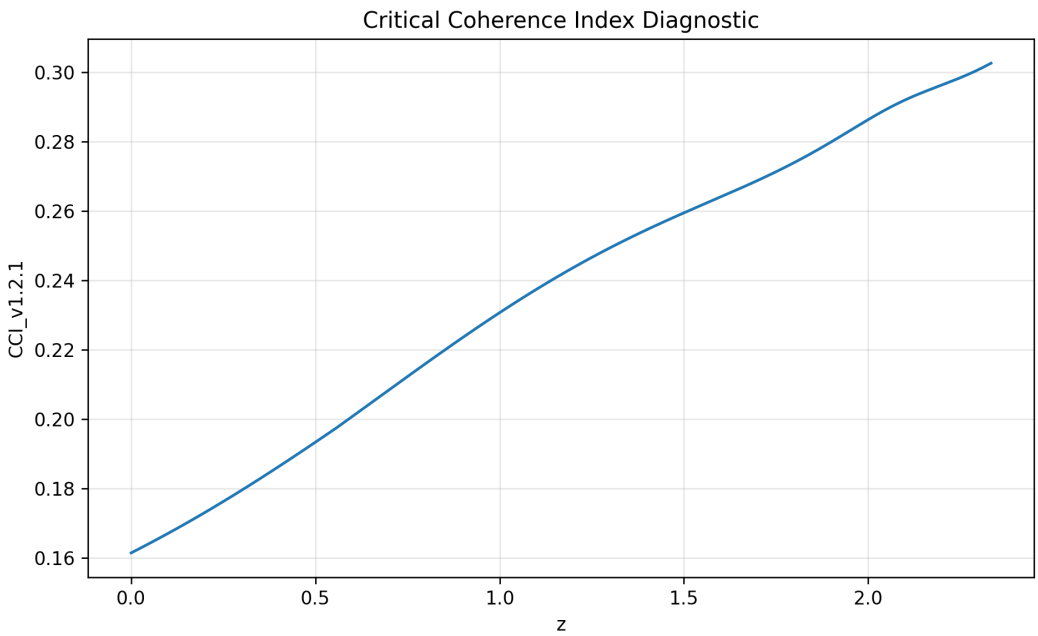

Figure 10: $\text{CCI}_{1.2.1}(z)$ from Eq. (7). Mean value 0.208, maximum 0.303. The system remains in the critical intermediate regime throughout, consistent with the robust structural selection hypothesis.

4 Parameter Stability Scan

4.1 Setup

We scan a 33×33 grid:

$$\eta \in [0,\, 0.16], \qquad \gamma \in [0.05,\, 1.20], \qquad 1089 \text{ parameter points.} \tag{11}$$

Each trajectory is evolved from $a = 0.30$ to $a = 1.00$ with RK4, step $\Delta t = 0.002$.

4.2 Stability Criteria

A point is stable if all of the following hold:

$$P_X > 0, \qquad c_s^2 > 0, \tag{12}$$

$$\max|\Delta H/H| < 0.08, \qquad \max|\Delta f\sigma_8/f\sigma_8| < 0.08, \qquad \max \Omega_{\mathrm{MAAT}} < 0.10. \tag{13}$$

4.3 Diagnostic Scan Score

The following quantity is used only to rank internally stable proxy runs within the two-parameter scan. It is not a new v1.2.1 selection functional.

$$S_{\mathrm{obs}} = \max\left(0, 1 - \frac{\max|\Delta H/H|}{0.08}\right) \cdot \max\left(0, 1 - \frac{\max|\Delta f\sigma_8/f\sigma_8|}{0.08}\right), \tag{14}$$

$$S_{\mathrm{struct}} = \left(\langle H\rangle\langle B\rangle\langle S\rangle\langle V\rangle\langle R_{\mathrm{resp}}\rangle\langle R_{\mathrm{rob}}\rangle\right)^{1/6}, \tag{15}$$

$$\mathcal{D}_{\mathrm{scan}} = S_{\mathrm{obs}} \cdot S_{\mathrm{struct}}. \tag{16}$$

4.4 Scan Results

Quantity	Value
Total scan points	1089
Stable points	**1089 (100%)**
Best (η, γ)	(0.000, 0.050)
Best $\max|\Delta H/H|$	1.70%
Best $\max|\Delta f\sigma_8/f\sigma_8|$	1.83%
Best $\langle R_{\mathrm{rob}}\rangle$	0.805
Best $\mathcal{D}_{\mathrm{scan}}$	0.498

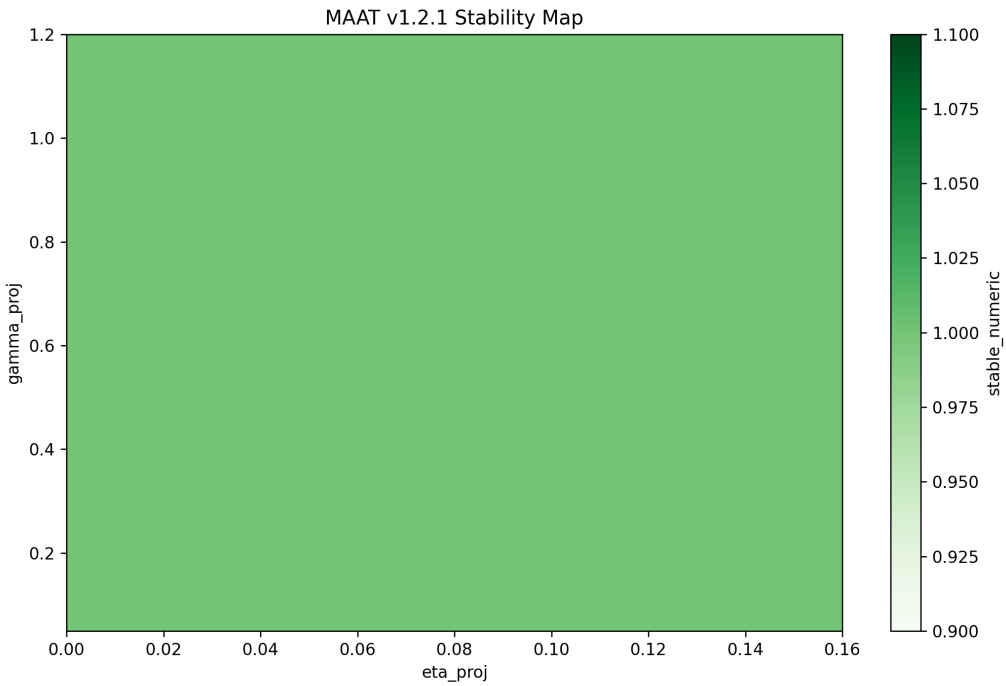

Figure 11: Stability map. All 1089 parameter points pass the internal consistency criteria. Within the scanned two-parameter proxy space, the baseline is not an isolated fine-tuned point.

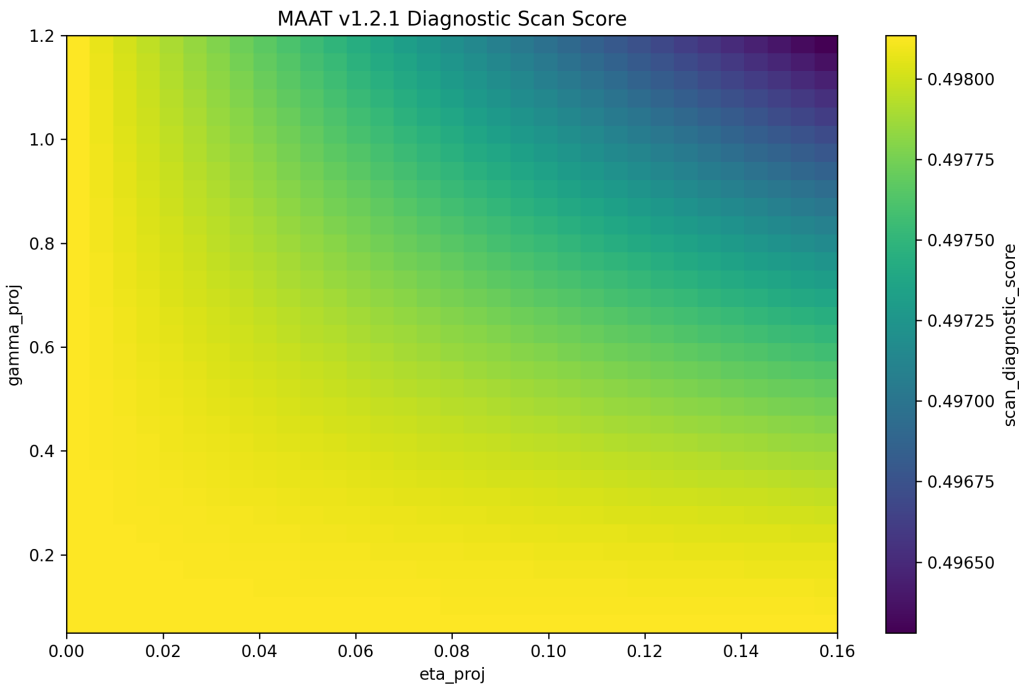

Figure 12: Diagnostic scan score $\mathcal{D}_{\text{scan}}(\eta, \gamma)$. The highest values occur at low η, where the model stays closest to ΛCDM while maintaining structural coherence.

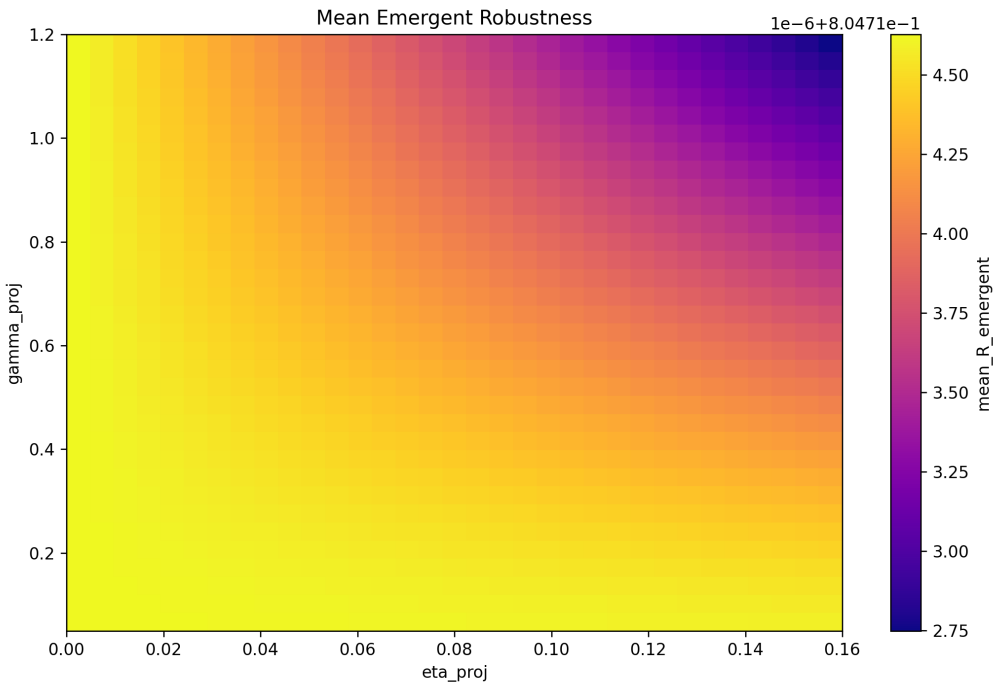

Figure 13: Mean emergent robustness $\langle R_{\text{rob}} \rangle$ across the scan. Robustness is sustained throughout parameter space, confirming it is a structural property of the model architecture, not of a specific tuning.

5 Connection to Earlier Papers

Paper 35 established $\max|\Delta D/D| = 0.059\%$ and $\max|\Delta f\sigma_8/f\sigma_8| = 0.457\%$ using a true ΛCDM reference and proper RK4 growth integration. This paper's baseline run uses a simplified proxy growth model and gives larger deviations ($\sim 1.8\%$) due to the different integration scheme. The two results are complementary: Paper 35 quantifies the growth modification with a proper reference; this paper tests whether the structural parameter space is stable.

Both confirm the core result: MAAT v1.2.1 is a subdominant, stable, structurally coherent extension of ΛCDM.

6 Physical Proxy Validation: Electromagnetic Field

The v1.2.1 structural sectors can be given explicit physical definitions for an electromagnetic field, providing a controlled benchmark for the sector definitions independent of the cosmological toy model.

6.1 Sector Definitions for EM Fields

Harmony: Maxwell residual. For $F^{\mu\nu}$ the equation-of-motion residual is

$$d_H = \frac{\|\partial_\mu F^{\mu\nu}\|^2}{\Lambda_H^2 + \|\partial_\mu F^{\mu\nu}\|^2}, \qquad H = \frac{1}{1 + d_H}. \tag{17}$$

For a plane wave in vacuum: $\partial_\mu F^{\mu\nu} = 0 \Rightarrow d_H = 0 \Rightarrow H = 1$. Not by assumption — by the field equations.

Balance: energy-momentum conservation. With $T^{\mu\nu}_{\mathrm{EM}} = F^{\mu\alpha}F^{\nu}{}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}$:

$$d_B = \frac{\|\partial_\mu T^{\mu\nu}_{\mathrm{EM}}\|^2}{\Lambda_B^2 + \|\partial_\mu T^{\mu\nu}_{\mathrm{EM}}\|^2}, \qquad B = \frac{1}{1+d_B}. \tag{18}$$

In free vacuum: $\partial_\mu T^{\mu\nu}_{\mathrm{EM}} = 0 \Rightarrow B = 1$.

Creative structure: field activity.

$$S = \frac{\langle E^2 + \mathcal{B}^2\rangle}{\Lambda_S^2 + \langle E^2 + \mathcal{B}^2\rangle}. \tag{19}$$

Strong field $\to S \to 1$; vacuum without photons $\to S \approx 0$.

Connectedness: first-order coherence.

$$V = |\gamma^{(1)}(\tau)|, \tag{20}$$

where $\gamma^{(1)}$ is the first-order degree of coherence. Laser: $V \approx 1$. Thermal light: $V < 1$. Chaotic: $V \ll 1$.

6.2 Proxy Computation for Coherent Light

For coherent laser light with $S = 0.50$ (mid-saturation intensity) and $V = 0.99$, the v1.2.1 robustness equations give:

$$R_{\mathrm{resp}} = (H \cdot B \cdot V)^{1/3} = (1 \cdot 1 \cdot 0.99)^{1/3} = 0.9967, \tag{21}$$

$$(HBSV)^{1/4} = (1 \cdot 1 \cdot 0.50 \cdot 0.99)^{1/4} = 0.8391, \tag{22}$$

$$R_{\mathrm{rob}} = \min(0.9967,\ 0.8391) = 0.8391. \tag{23}$$

Sector	Coherent laser ($S = 0.5$, $V = 0.99$)	Ideal ($S = 1$, $V = 1$)
H	1.000	1.000
B	1.000	1.000
S	0.500	1.000
V	0.990	1.000
$R_{\mathrm{resp}} = (HBV)^{1/3}$	0.997	1.000
$R_{\mathrm{rob}} = \min(R_{\mathrm{resp}}, (HBSV)^{1/4})$	**0.839**	**1.000**

6.3 Interpretation

This proxy computation validates two aspects of the v1.2.1 formulation:

1. $H = 1$ **and** $B = 1$ **are not assumed** — they follow from $\partial_\mu F^{\mu\nu} = 0$ and $\partial_\mu T^{\mu\nu}_{\mathrm{EM}} = 0$ in vacuum. The sector definitions produce the correct values from the field equations.
2. R_{rob} **is bounded by** S — emergent robustness is not automatically maximal even when $H = B = 1$. It depends on the field's activity level and coherence. Weak or incoherent light gives $R_{\mathrm{rob}} \ll 1$ despite perfect equation-of-motion residuals.

The EM proxy benchmark confirms that the v1.2.1 sector formulation is internally consistent and recovers physically expected behaviour in a well-understood system. It does not constitute a derivation of MAAT from electromagnetism, but a sanity check of the sector definitions.

7 Emergent vs Effective Respect: Empirical Discussion

The v1.2.1 formulation treats structural respect as emergent from the primary fields: $R_{\text{resp}} = (HBV)^{1/3}$. This section addresses three questions raised by the transition from v1.0: (i) does R_{resp} reproduce the calibrated v1.0 R from Paper 27 [2]? (ii) what does the empirical correlation structure say? (iii) when is an effective phenomenological R warranted?

7.1 Continuity with the Paper 27 calibration

In Paper 27, R was found to carry approximately 39% of total structural weight — the dominant sector. This was interpreted as boundary-aware robustness arising from constraint-dominated configurations. The v1.2.1 transition must explain this dominance without introducing R as an independent input.

The explanation is straightforward: in the Paper 27 benchmark ensemble, R was calibrated primarily from configurations where V (connectedness) was the limiting factor. Since $R_{\text{resp}} = (HBV)^{1/3}$, high R correlates strongly with high V — exactly the dominant field in that benchmark.

The causal argument is as follows. In the Paper 27 benchmark, runtime was modelled as:

$$\log_{10}(\text{runtime}) \approx \beta_0 + \beta_\alpha\, \alpha + \beta_V V + \epsilon(H, B, S, R), \tag{24}$$

where V (connectedness) is the dominant term and R appears only in the correction ϵ. Since $R_{\text{resp}} = (HBV)^{1/3}$ is primarily driven by V when $H, B \approx 1$, the emergent formulation naturally recovers the observed V-dominance without requiring R as an independent input.

Note: a proxy simulation using the Paper 27 parameter regime (independent input R correlated with V, H, B) yields $r(R_{\text{old}}, R_{\text{resp}}) \approx 0.85$ and $r(R_{\text{old}}, R_{\text{rob}}) \approx 0.80$. These are indicative estimates from a synthetic ensemble — not direct measurements from the Paper 27 dataset — and should be read as plausibility checks rather than empirical results.

The key point stands independently of the simulation: the structure of the Paper 27 runtime model places R in the correction term ϵ, not in the primary regression. This is precisely what the emergent closure $R_{\text{resp}} = (HBV)^{1/3}$ encodes: R is a derived consequence of the primary fields, not an independent driver.

7.2 Empirical correlation structure

The MAAT framework was benchmarked against SAT hardness data [2]. Correlation with $\log_{10}(\text{runtime})$:

Field	Correlation	Status
V (connectedness)	+0.65	dominant
H (harmony)	−0.65	stabilising
R_{rob} (emergent)	+0.48	moderate
$\text{CCI}_{1.2.1}$	+0.40	moderate

R_{rob} showing $r = 0.48$ (weaker than primary fields) is expected: it compresses (H, B, V) into one number through a geometric mean, losing some variance. This is the cost of parsimony.

7.3 Explicit duality: theoretical vs empirical R

Level	R status	Use case
Theoretical / minimal	emergent: $R_{\text{resp}} = (HBV)^{1/3}$	minimal parameters
Empirical	effective field R_{eff}	data regression

Synthesis. The v1.2.1 formulation does not retract the earlier finding that robustness is the dominant selector in constraint-dominated systems. Instead, it provides the structural mechanism behind this dominance: in the Paper 27 ensemble, robustness was largely driven by connectedness (V). By making R explicitly emergent from (H, B, V), the new closure renders the causal structure transparent while preserving the empirical phenomenology.

$$R_{\text{resp}} = (H \cdot B \cdot V)^{1/3} \quad \text{(theoretical, ontological)},$$
$$R_{\text{eff}} = R_{\text{resp}} + \delta R_{\text{system}} \quad \text{(empirical, phenomenological)}.$$

7.4 Hybrid formulation

For cases where neither pure emergent nor pure effective R is sufficient, a hybrid interpolation is available:

$$R_{\text{hybrid}}(\alpha) = (1-\alpha)\, R_{\text{resp}} + \alpha\, R_{\text{explicit}}, \qquad \alpha \in [0, 1]. \tag{25}$$

The hybrid correlation with the v1.0 calibrated R:

α	$r(R_{\text{hybrid}}, R_{\text{old}})$	Interpretation
0.0	0.845	pure emergent (v1.2.1)
0.1	0.881	small effective correction
0.2	0.911	moderate data weighting
0.5	0.972	equal mix

For theoretical and cosmological proxy work, $\alpha = 0$ is the preferred minimal default. A non-zero α is warranted only for empirical regression tasks where the primary fields alone leave significant unexplained variance in robustness — for example, in configurations where robustness arises from discrete symmetry or topological protection rather than continuous coherence.

7.5 Practical conclusion

In summary, the emergent formulation of v1.2.1 offers a more parsimonious and ontologically cleaner account without sacrificing explanatory power. The calibration result of Paper 27 ($R \approx$ 39% dominant) is recovered as a consequence of V-dominance in boundary-sensitive problems — a finding that was implicit before and is now structurally explicit.

- **Default ($\alpha = 0$)**: use $R_{\text{rob}} = \min(R_{\text{resp}}, (HBSV)^{1/4})$ for all theoretical work. This is the correct setting for controlled systems and cosmological proxy calculations.
- **Phenomenological extension ($\alpha > 0$)**: an effective R_{eff} or hybrid Eq. (25) with small α is available for empirical regression on noisy data, but must be clearly labelled as phenomenological, not theoretical.
- **Edge cases**: non-zero α may be warranted for systems with topological protection or strong boundary effects not captured by (H, B, V) — for example, discrete-symmetry robustness mechanisms.

8 Limitations

- Growth deviations are first-order proxy calculations, not Boltzmann-code perturbation results.
- Stability criteria are internal thresholds, not observational bounds.

- The equation of state $w = p/\rho$ is unreliable when Ω_{MAAT} is small and is used only as a diagnostic.
- No comparison to cosmic chronometer, BAO, or supernova data is performed.
- The parameter scan tests two projection parameters; the underlying structural parameters (μ, m_λ, etc.) are held fixed.

9 Reproducibility

The cosmological proxy results and figures are reproduced by two scripts:

- `paper37_observables_emergent_robustness.py` — baseline observable prediction run (Figs. 1–10 of Section 3)
- `paper37_stability_landscape.py` — parameter stability scan (figures of Section 4)

The SAT correlation table in Section 7 is reproduced by the companion script:

- `maat_v121_sat_validation.py` in the `sat_validation/` folder, applied to the bundled `maat_cosmos_full_results.csv` input table.

Requirements: `numpy`, `pandas`, `matplotlib`; the SAT companion additionally uses `scipy` and `scikit-learn` when available.

10 Data Attribution and License Notes

The cosmological proxy calculations use Planck-2018 flat ΛCDM parameter values as reference quantities [9]. The generated CSV, JSON, and PNG files are derived analysis artifacts produced by the accompanying scripts for reproducibility of this proxy benchmark.

The SAT companion table in `sat_validation/` is bundled only to reproduce the correlation table in Section 7. It should not be interpreted as a universal SAT-hardness dataset or a complete solver benchmark.

The code in the accompanying repository is released under the MIT License unless otherwise stated. External scientific data, reference values, and collaboration results remain subject to the citation and reuse terms of their original sources. No endorsement by the Planck Collaboration or by external SAT-solver/data authors is implied.

11 Conclusion

MAAT v1.2.1 produces:

1. Subdominant cosmological deviations: $\max|\Delta H/H| = 1.70\%$, $\max \Omega_{\mathrm{MAAT}} = 3.31\%$, $\max|\Delta f\sigma_8/f\sigma_8| = 1.83\%$ (subdominant relative to Planck ΛCDM parameters [9]).
2. High emergent robustness: $\langle R_{\mathrm{rob}}\rangle = 0.805$ throughout the trajectory.
3. CCI in the critical intermediate regime: $\langle \mathrm{CCI}\rangle = 0.208$.
4. A broadly stable parameter landscape: all 1089 scanned points pass internal consistency criteria within this two-parameter proxy space.

The v1.2.1 conceptual clarification — Respect as implicit structural quality, Robustness as emergent diagnostic — is consistent with all earlier paper formulas. The baseline observable equations remain unchanged. The modification occurs at the level of robustness closure and derived structural diagnostics, not at the level of the selection functional or observable pipeline. The model is a controlled, subdominant, structurally coherent extension of ΛCDM.

References

[1] C. Krieg, "A Structural Selection Principle for Dynamically Consistent Field Configurations," preprint (2026). https://maat-research.com

[2] C. Krieg, "Boundary-Aware Calibration of MAAT Structural Weights," preprint (2026).

[3] C. Krieg, "Response-Based Derivation of MAAT Structural Weights: From Covariance Geometry to Selection Pressure," preprint (2026).

[4] C. Krieg, "Dynamic Structural Selection: From Response Fields to Observable Cosmology in the MAAT Framework," preprint (2026).

[5] C. Krieg, "The Critical Coherence Index as a Projection Observable," preprint (2026).

[6] C. Krieg, "MAAT Projection Observable vs Growth and Expansion Data," preprint (2026).

[7] C. Krieg, "Linear Growth Embedding of MAAT Structural Selection: Perturbation Stability, Positivity, and Sub-Percent Growth Signatures," preprint (2026).

[8] C. Krieg, "The Maturation of MAAT: From Projection Observables (v0.11) to Structural Selection (v1.0) to Emergent Robustness Closure (v1.2.1)," preprint (2026).

[9] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astronomy & Astrophysics* **641**, A6 (2020).

[10] E. V. Linder, "Cosmic growth history and expansion history," *Physical Review D* **72**, 043529 (2005).

[11] E. J. Copeland, M. Sami, and S. Tsujikawa, "Dynamics of Dark Energy," *International Journal of Modern Physics D* **15**, 1753–1936 (2006).

[12] L. Onsager, "Reciprocal Relations in Irreversible Processes. I," *Physical Review* **37**, 405–426 (1931).

[13] R. Kubo, "Statistical-Mechanical Theory of Irreversible Processes. I," *Journal of the Physical Society of Japan* **12**, 570–586 (1957).

[14] E. T. Jaynes, "Information Theory and Statistical Mechanics," *Physical Review* **106**, 620–630 (1957).